Name: _____

WORKSHEET

Addition of vectors using components

1 Evaluate the following.

a $(1, 3) + (2, 5)$

b $(5, 1) + (8, 4)$

c $(-2, 9) + (7, 3)$

d $(-8, -5) + (9, 4)$

e $(1, 0) + (-3, -7)$

f $(-5, -2) + (-6, -10)$

g $(0, 11) - (6, -4)$

h $(-3, 7) - (-1, 5)$

i $(1, 1) + (2, 2) + (3, 3)$

j $(2, -5) + (9, 3) + (-6, 4)$

2 Convert to component form and evaluate.

a $(2, 90°) + (5, 0°)$

b $(4, 180°) + (8, 90°)$

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c $(12, 270°) + (3, 180°)$

d $(7, 0°) + (9, 270°)$

e $(5, 100°) + (3, 45°)$

f $(11, 160°) + (4, 189°)$

g $(0.5, 250°) + (2, 300°)$

h $(4.7, 216°) + (9.8, 333°)$

i $(10, 10°) + (5, 305°)$

j $(r_1, \theta_1) + (r_2, \theta_2)$

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3 Convert to component form and evaluate.

a $\mathbf{i} + \mathbf{j}$

b $[2\mathbf{i} + 5\mathbf{j}]$

c $8\mathbf{i} + 3\mathbf{j}$

d $6\mathbf{i} + \mathbf{j}$

e $0\mathbf{i} + 9\mathbf{j}$

f $2\mathbf{i} - 3\mathbf{j}$

g $5\mathbf{i} - 7\mathbf{j}$

h $[3\mathbf{i} - 8\mathbf{j}]$

i $-6\mathbf{i} - 9\mathbf{j}$

j $-5\mathbf{i} - 0\mathbf{j}$

4 Given $\mathbf{u} = (3, 7)$ $\mathbf{v} = (5, 210°)$ and $\mathbf{w} = 9\mathbf{i} - 4\mathbf{j}$ evaluate.

a $\mathbf{u} + \mathbf{v}$

b $\mathbf{u} + \mathbf{w}$

c $\mathbf{u} + \mathbf{v} + \mathbf{w}$

d $3\mathbf{u} - 2\mathbf{v} - 4\mathbf{w}$

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e $-5\mathbf{u} + 9\mathbf{v} - 2\mathbf{w}$

f $8\mathbf{u} - 2\mathbf{v} + 8\mathbf{w}$

g $0\mathbf{u} + 3\mathbf{v} + 7\mathbf{w}$

h $-6\mathbf{u} - 5\mathbf{v} + 2\mathbf{w}$

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Answers

1 **a** $(3, 8)$

b $(13, 5)$

c $(5, 12)$

d $(1, -1)$

e $(-2, -7)$

f $(-11, -12)$

g $(-6, 15)$

h $(-2, 2)$

i $(6, 6)$

j $(5, 2)$

2 **a** $(5, 2)$

b $(-4, 8)$

c $(-3, -12)$

d $(7, -9)$

e $(1.25, 7.05)$

f $(-14.29, 3.14)$

g $(0.83, -2.20)$

h $(4.93, -7.21)$

i $(12.72, -2.36)$

j $(r_1, \cos\theta_1 + r_2\cos\theta_2, r_1\sin\theta_1 + r_2\sin\theta_2)$

3 **a** $(1, 1)$

b $(2, 5)$

c $(8, 3)$

d $(6, 1)$

e $(0, 9)$

f $(2, -3)$

g $(5, -7)$

h $(3, -8)$

i $(-6, -9)$

j $(-5, 0)$

4 **a** $(-1.33, 4.5)$

b $(12, 3)$

c $(7.67, 0.5)$

d $(-18.34, 42)$

e $(-71.97, -49.5)$

f $(104.66, 29)$

g $(50.01, -35.5)$

h $(21.65, -37.5)$

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